Single Agent Systems & Agent Pipelines - Quiz

Instructions:
Answer the following questions. These questions are designed to test your understanding of agent systems, tools, and evaluation.

1. Explain the concept of a stateful directed graph in agent pipelines. How does it differ from a simple linear pipeline?

A stateful directed graph is a workflow where the agent remembers information from previous steps and can move between different paths depending on the situation, whereas a simple linear pipeline always follows one fixed sequence from start to end. In a stateful graph, the agent can make decisions, go back if needed or follow different routes based on the input.

2. Describe the role of nodes and edges in an agent workflow. Give an example of each.

Nodes are the individual tasks or actions that the agent performs.
Edges are the connections between these tasks and decide what happens next.
Example:- a calculator tool is a node and the connection from checking the user query to opening the calculator tool is an edge.

3. What is conditional routing in an agent system? Design a simple rule-based routing logic for three different query types.

Conditional routing in an agent system means sending a user's query to the correct tool based on its type.

- If the query contains "calculate", send it to the calculator tool.
- If the query contains "keywords", send it to the keyword extraction tool.
- Otherwise, treat it as a general question and return a normal response.

4. Why are cycles (loops) important in agent pipelines? Provide a use case where a retry loop is necessary.

Loops allow the agent to repeat a task until it succeeds or gets the correct result. For example, if an API call fails because of a network problem, the agent can retry a few times instead of stopping immediately.

5. Explain how a single-agent system can simulate multi-agent behavior internally.

A single-agent system can behave like multi-agent by using different tools for different tasks. It first understands the user's request, decides which tool is needed, runs that tool, and then gives the final answer. Although only one agent is used, it performs different roles internally.

6. What are JSON schema tools? How do they help in structuring tool inputs and outputs?

JSON schema tools define a fixed format for data. They make sure that the input and output follow the same structure every time. This reduces errors and makes it easier for different tools to communicate with each other.

7. Compare sequential tool calls and parallel tool calls. When would you prefer one over the other?

Sequential tool calls run one after another so that each step can use the result from the previous one. Parallel tool calls run multiple tools at the same time. I would use sequential calls when the tasks depend on each other and parallel calls when the tasks are independent and can be completed together to save time.

8. How would you implement error handling in a tool-using agent? Provide at least two strategies.

Two simple error handling methods are:

1. Use try-except blocks to catch errors and stop the program from crashing.
2. Return a clear error message or retry the operation if something goes wrong.

9. What is trajectory evaluation in agent systems? Why is it important beyond just checking final output?

Trajectory evaluation means checking all the steps the agent followed before giving the final answer. It is important because even if the final answer is correct, the agent may have taken unnecessary or incorrect steps. Evaluating the whole process helps improve the system.

10. Define task completion rate and cost metrics. How would you measure and optimize them in a real-world system?

Task completion rate is the percentage of tasks the agent finishes successfully.
Cost metrics measure the resources used such as time, API calls, or computing power.
In a real-world system, I would measure how many tasks are completed correctly and how much time or cost is involved and to improve performance, I would reduce unnecessary steps, optimize the workflow, and use the right tools for each task.